DP Biology – C4.1

Populations and communities

- How do interactions between organisms keep population sizes in check?
- Which interactions make the populations in a community depend on one another?

SL statements C4.1.1–18. Textbook pp. 548–577. There is no HL content in C4.1.

Image: lesser flamingos, Lake Nakuru, Kenya. Syllabub, CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C4.1.1–8	Estimate population size with quadrats and the Lincoln index, and explain carrying capacity and sigmoid growth
2	C4.1.9–14	Describe communities, intraspecific and interspecific relationships, invasive species and tests for competition
3	C4.1.15–18	Use the chi-squared test, and explain predator-prey cycles, control from above and below, and allelopathy

The big picture: counting and connecting



Populations grow until **resources** and **interactions** (competition, predation, disease) hold them in check.

Big question

Why do lynx numbers rise and fall about every ten years?

What is a population?



Lesser flamingos, Lake Nakuru. Syllabub, CC BY-SA 3.0.

A **population** is a group of one species in the same area at the same time, normally interbreeding.

Populations are separated by **reproductive isolation**: distance, mountains or water.

Discuss

Are flamingos at a lake 300 km away the same population? What would you need to know?

Estimating population size

C4.1.2

Counting every individual is usually impossible, so we count a **random sample** and scale up.

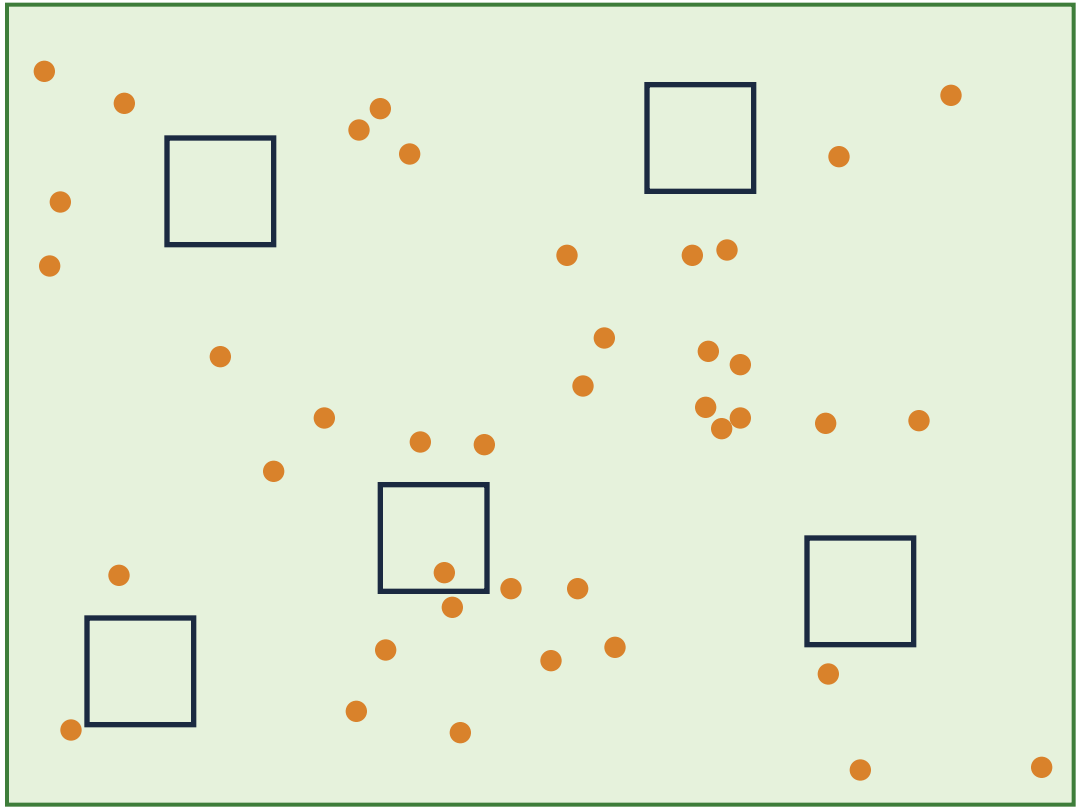
Random sampling avoids **bias**.

Organisms	Method
Sessile (plants, barnacles)	Random quadrats
Motile (beetles, voles)	Capture-mark-release-recapture

Key idea

An estimate is never exact. More samples give a more reliable estimate.

Random quadrat sampling



random coordinates from two tapes at right angles

Scale up

Population = mean per quadrat × (total area / quadrat area).



A 1 m quadrat on a rocky shore. M.-N. De Casamajor, Ifremer, CC BY 4.0.

Worked example: daisies

C4.1.3

Quadrat	1	2	3	4	5	6	7	8	9	10
Daisies	3	5	0	4	2	6	1	4	3	2

1. Field 600 m^2; quadrat 1 m^2
2. Mean = 30 / 10 = 3.0 per m^2
3. Estimate = 3.0 × 600 = 1800
4. SD = 1.8 (calculator)

Standard deviation

Shows spread around the mean: large = patchy, small = even. Use a calculator.

Try it

A second student gets 2.4 per m^2. Give two reasons and one improvement.

Capture-mark-release-recapture

C4.1.4



Worked example

40 marked; 50 caught later, 10 marked. Population = 40 × 50 / 10 = 200.

Assumption	Why
Marks not lost	Else overestimate
Full mixing	Random sample
No births, deaths, migration	Same population

Lincoln index

Population size = (M × N) / R.

Practice: a sea turtle population

C4.1.4



Green turtle grazing on seagrass. P. Lindgren, CC BY-SA 3.0.

Year	M	N	R
1	30	36	12
2	28	40	14
3	25	30	15

Invented values, for practice.

Activity

Estimate each year's population and describe the trend. Turtles live long and swim far: which assumption is most likely broken?

Carrying capacity and density-dependent control

C4.1.5–6

Carrying capacity (K) is the largest population an environment can support. Resources limit it: food, water, light, space.

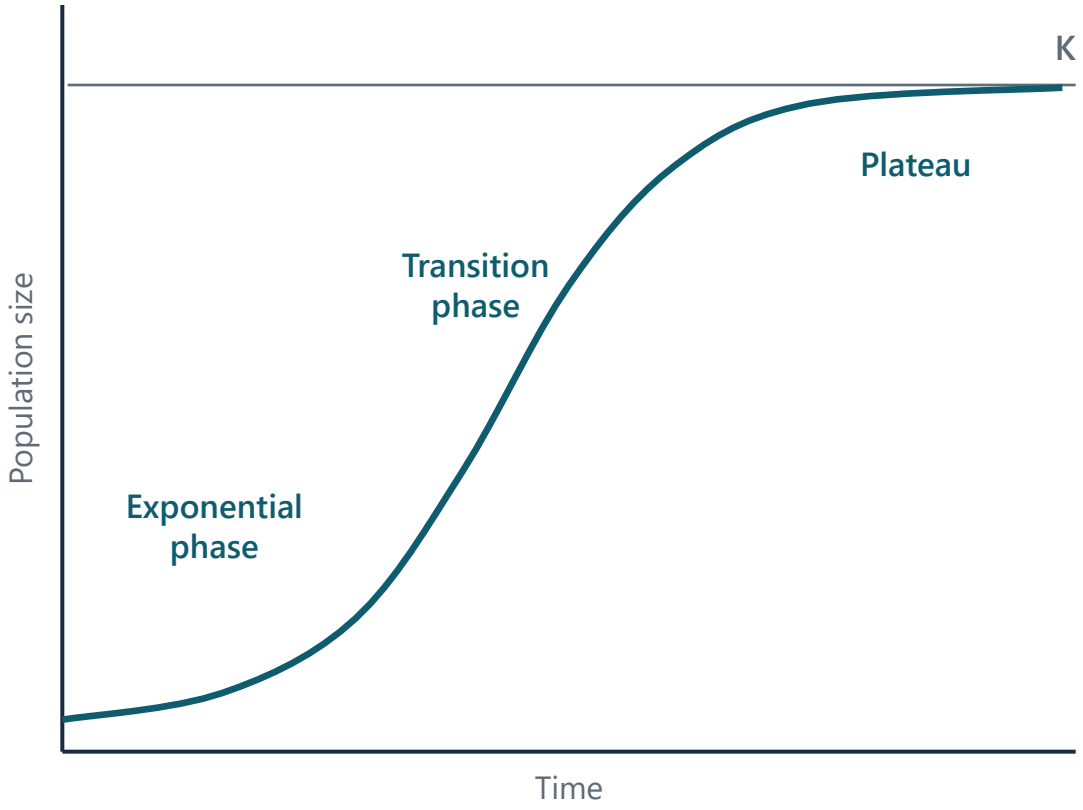
Density-dependent factor	Stronger when crowded because…
Competition	Less resource per individual
Predation	Prey easier to find
Disease	Spreads faster between neighbours

Negative feedback

Above K, deaths rise and births fall; below K, the reverse. Floods and fires are density-independent.

The sigmoid population growth curve

C4.1.7



Exponential: plenty of resources.

Transition: density-dependent factors increase.

Plateau: births = deaths at K.

Example

Bearded vultures released in the Alps since 1986 are now growing fast: early exponential phase.

Modelling sigmoid growth with duckweed

C4.1.8



Lemna minor. Karelj, CC BY-SA 3.0.

1. Put a few duckweed plants in nutrient solution in light
2. Count fronds every day or two
3. Plot fronds against time

Test the model

Plot fronds on a log scale: exponential growth gives a straight line; levelling shows K.

Extension

Change nutrient concentration or container area; predict the effect on K.

Check-in: class 1

C4.1.1–8

1. Define population.
2. Explain why samples must be random.
3. 40 marked, 50 caught later, 8 marked: estimate the population.
4. Define carrying capacity.
5. Describe the three phases of sigmoid growth.

How to use this

Answer without your notes, then check against slides 3–12.

A community: all the interacting organisms

C4.1.9

A **community** is all the populations of different species in an area, including fungi and bacteria.

With the abiotic environment it forms an **ecosystem**.



A coral reef community. Jim Maragos, USFWS, public domain.

Activity

List ten organisms in a local community and draw lines showing how they interact.

Competition and cooperation within a species

C4.1.10



Spotted hyenas. S. Jurvetson, CC BY 2.0.



Leafcutter ants. Kathy and Sam, CC BY 2.0.

Interaction	Examples
Competition	Wildebeest males fight for territory; seedlings compete for light
Cooperation	Hyenas hunt in groups; leafcutter ants farm fungus; meerkats share care of young

Discuss

Why is competition never removed completely, even in a cooperative group?

Relationships between species

C4.1.11

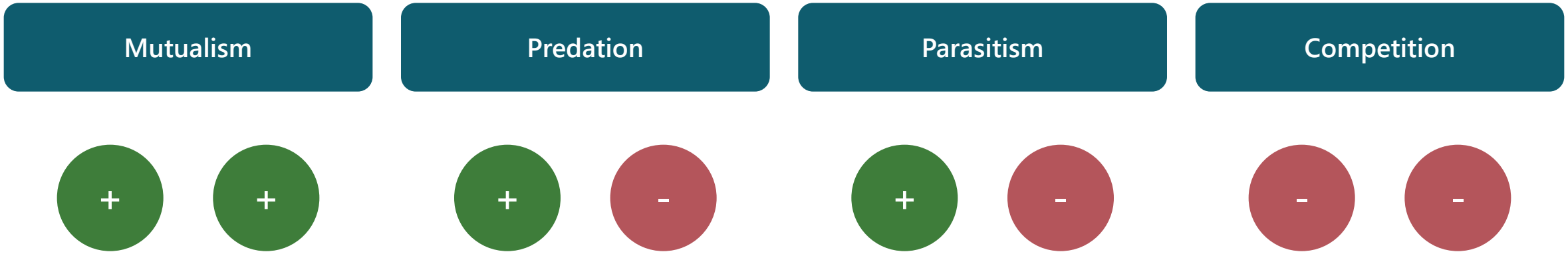
Relationship	What happens	Example
Herbivory	Consumer eats a producer	Bison and grass
Predation	Predator kills and eats prey	Lynx and hare
Competition	Same limited resource	Lions and leopards
Mutualism	Both benefit	Rhizobium in clover
Parasitism	Parasite harms host	Ticks on deer
Pathogenicity	Pathogen causes disease	Chytrid fungus in frogs

Activity

Find a local example of each.

Who wins, who loses?

C4.1.11

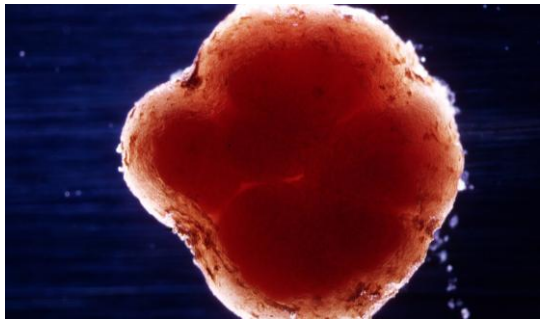


Read it

Green = benefits; red = harmed. Herbivory is like predation (+/-), though the plant often survives.

Mutualism: both species benefit

C4.1.12



Root nodule. CSIRO, CC BY 3.0.



Lady's slipper orchid. Johan N, public domain.



Coral reef. J. Maragos, public domain.

Partners	Host gains	Partner gains
Legume + Rhizobium	Fixed nitrogen (ammonium)	Sugars; low-oxygen nodule
Orchid + fungus	Sugars, minerals, water for seeds	Later, carbon compounds
Coral + zooxanthellae	Sugars, oxygen	Shelter, light, CO_2

Endemic and invasive species

C4.1.13



Red squirrel. Jojo, CC BY-SA 3.0.

Endemic species occur naturally; **invasive** ones were introduced by humans and spread.

Invaders often lack their native predators and pathogens.



Grey squirrel. PierreSelim, CC BY 3.0.

Example

Grey squirrels eat acorns better, store more fat and carry squirrel pox. Red squirrels have vanished from most of England and Wales.

Invasive species in Switzerland



Nutria. G. Chernilevsky, public domain.



Himalayan balsam. R. Flogaus-Faust, CC BY 4.0.

Species	Advantage
Nutria	Eats waterside plants; burrows into banks
Himalayan balsam	2 m in one season; shades natives; exploding pods
Japanese knotweed	Spreads by rhizomes

Activity

Explain how one invader outcompetes a native species for one resource.

Testing for interspecific competition

C4.1.14

Approach	Method	Example
Laboratory	Grow alone and together	Two Paramecium species
Field observation	Do they co-occur less than expected?	Chi-squared (C4.1.15)
Field manipulation	Remove one species from plots	Connell's barnacles

Careful
Better growth without the other species suggests competition but does not prove it: they might just need different conditions.

Check-in: class 2

C4.1.9–14

1. Define community.
2. Give an example of intraspecific cooperation.
3. Distinguish parasitism and mutualism.
4. Explain why invasive species often spread.
5. Give three ways to test for competition.

How to use this

Answer without your notes, then check against slides 14–21.

Chi-squared: are two species associated?

C4.1.15



Ling heather. jensu, CC0.



Bracken. L. A. Ratledge, CC BY 4.0.

Record **presence or absence** in many random quadrats. H0: distributed independently. H1: associated.

Observed (100)	Bracken +	Bracken -	Total
Ling +	18	42	60
Ling -	32	8	40
Total	50	50	100

Invented values, for practice.

Calculating chi-squared

Expected	Bracken +	Bracken -
Ling +	30	30
Ling -	20	20

1. Expected = row × column total / grand total
2. Sum $(O - E)^2 / E = 4.8 + 4.8 + 7.2 + 7.2 = 24.0$
3. df = 1; critical value 3.84 (p = 0.05)

Conclusion

24.0 > 3.84: reject H0. A negative association: they co-occur less than expected. Could be competition, allelopathy or different soil needs: the test shows association, not cause.

Practice: two seaweeds

C4.1.15

Observed (80)	P +	P -	Total
Q +	24	8	32
Q -	16	32	48
Total	40	40	80

Invented values, for practice.

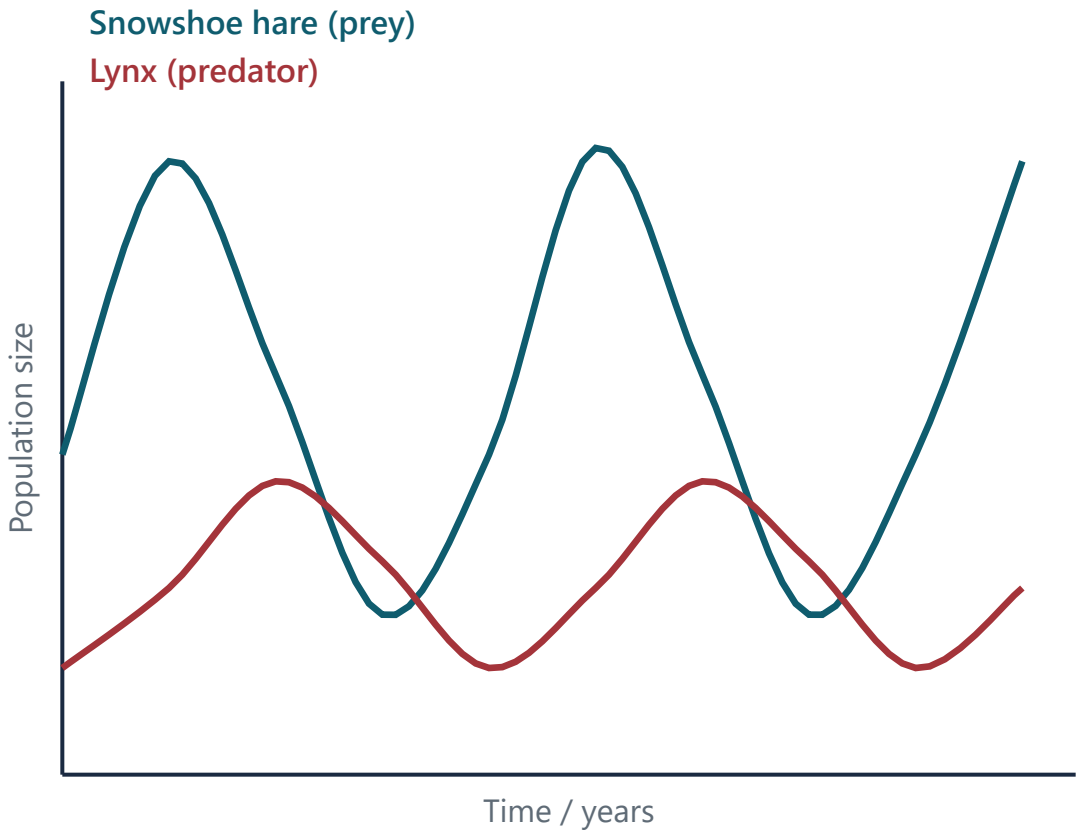
1. State H0 and H1
2. Calculate expected values
3. Calculate chi-squared
4. Compare with 3.84
5. Conclude

Extension

Plan a removal experiment to test whether the association is caused by an interaction.

Predator-prey cycles

C4.1.16



General shape only. Peaks about every 10 years.

1. More hares: lynx rise
2. More lynx: hares fall
3. Less food: lynx fall; hares recover



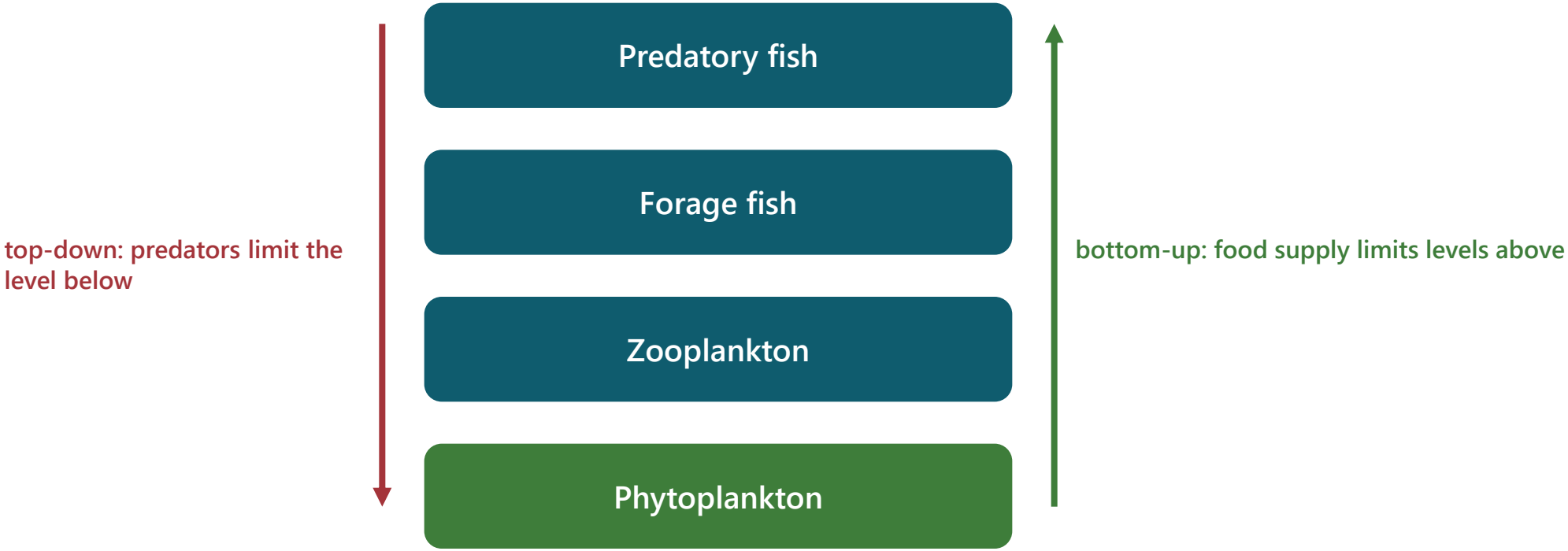
Lynx. USFWS, public domain.



Snowshoe hare. A. Schmierer, CC0.

Control from above and from below

C4.1.17



Key idea

Both act in a community, but usually one dominates. Removing wolves shows top-down control clearly.

Allelopathy and antibiotics

C4.1.18



Bracken. L. A. Ratledge, CC BY 4.0.



Penicillium colonies. Atalyssa, CC BY 4.0.

	Allelopathy	Antibiotics
What	Plant chemicals stop other plants	Microbe chemicals kill bacteria
Example	Bracken, walnut	Penicillium: penicillin (1928)
Effect	Fewer plant competitors	Fewer bacterial competitors

Similarity

Both release a chemical that deters competitors.

Working with data: predator removal

C4.1.16–17

Mean deer per km^2 and seedlings per m^2 before and after wolves returned to a valley. Invented values; realistic pattern.

Year	Deer / km^2	Seedlings / m^2
Before wolves	12	2
5 years after	7	6
10 years after	5	11

1. Describe both trends.
2. Name the type of control shown.
3. Suggest one other explanation.

Check-in: class 3

C4.1.15–18

1. State H0 for a chi-squared test of association.
2. State the critical value for df = 1, p = 0.05.
3. Explain predator-prey cycles.
4. Distinguish control from above and from below in a food chain.
5. Compare allelopathy and antibiotics.

How to use this

Answer without your notes, then check against slides 23–29.

Common mistakes to avoid

Not this	But this
K is the largest population ever reached.	It is the maximum supported long term.
Lincoln: divide by total caught.	Divide by marked recaptures.
A significant chi-squared proves cause.	It shows association.
Any non-native species is invasive.	Invasive species spread and harm.

Summary: populations and communities

C4.1.1–18

	Key idea
C4.1.1–4	Populations; quadrats; Lincoln index
C4.1.5–8	K, density dependence, sigmoid growth, duckweed
C4.1.9–12	Communities; intra- and interspecific relations; mutualism
C4.1.13–14	Invasive species; tests for competition
C4.1.15	Chi-squared association
C4.1.16–17	Predator-prey cycles; control from above or below
C4.1.18	Allelopathy and antibiotics

Check that you can...

1. **Estimate** population size
2. **Explain** carrying capacity and sigmoid growth
3. **Outline** interspecific relationships
4. **Apply** the chi-squared test
5. **Explain** predator-prey cycles and control
6. **Compare** allelopathy and antibiotics

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. 20 snails are marked and released. Later, 30 are caught, and 6 of them are marked. Estimate the population.

2. Explain why predation is a density-dependent factor.

3. State one benefit to each partner in the mutualism between legumes and Rhizobium.

Image credits

Large number of flamingos at Lake Nakuru.jpg – Syllabub – CC BY-SA 3.0

Quadrats sur le site Abbadia (Ifremer 00380-49168 – 1942).jpg – Marie-Noelle De Casamajor – CC BY 4.0

Green Sea Turtle grazing seagrass.jpg – P.Lindgren – CC BY-SA 3.0

Lemna minor Prague 2012 1.jpg – Karelj – CC BY-SA 3.0

Coral reef fish pacific blue tan paracanthurus hepatus.jpg – Jim Maragos, U.S. Fish and Wildlife Service – Public domain

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SNOWSHOE HARE (Lepus americanus) (5-28-2015) quoddy head, washington co, maine -02 (18988733919).jpg – ALAN SCHMIERER – CC0

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